



KRONOS FUSION ENERGY

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# *The Plasma Diagnostic Suite and Real-Time State Estimation for a Compact Spherical-Tokamak Breeder: Magnetics, Thomson, Interferometry, and Bolometry Fused by a Graph-Neural-Network Reconstructor*

An autonomous controller can act only on the state it can see, and in a compact spherical tokamak the state must be produced faster than the plant evolves, from a diagnostic set...

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THE OFFICIAL RECORD

What this document is, and how to read its numbers

This is a combined editorial. It gathers, in one volume, the two design studies Kronos Fusion Energy released in 2026 — a compact spherical-tokamak tritium/helium-3 breeder and a deuterium–helium-3 tandem-mirror generator — together with the intellectual-property estate, the people, and, in the internal edition, the full commercial case. The scientific text is the same text that appears in the arXiv preprints and the journal submissions; the editorial adds the apparatus a reader needs to hold the whole program at once, and nothing in the physics is altered to fit it.

THE HONESTY STANCE IS THE ASSET

Every headline quantity carries an **evidence class** — DERIVED RECOMPUTED MEASURED REQUIREMENT — and, where a number is a modelled extrapolation rather than a demonstrated value, it is labelled as such and its downside is shown next to it. Several quantities in the underlying corpus moved after first publication; where a value has been withdrawn or restated, the record says so plainly. This is deliberate: a diligence team will find the load-bearing assumptions, so the document surfaces them first.

TWO EDITIONS

This volume is issued in two editions. The **public edition** (light) carries the physics and the design only — no dollar, price, or per-kilo-gram figure appears anywhere in it. The **internal edition** (dark) adds the economics, the funding architecture, and the defense-commercial analysis, and is marked *Confidential* accordingly; it is not for public distribution. Where the commercial case rests on a modelled assumption rather than a demonstrated one, it is labelled as such and its downside is shown alongside it, on the same terms as the physics.

|                 |                                                                                                                                                                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TITLE           | The Kronos Fleet — Hyperion · Aegis · MetroVolt: A Combined Design & Commercialization Study                                                                                                                                                                       |
| AUTHOR          | Priyanca Ford, Founder & CEO, on behalf of Kronos Fusion Energy                                                                                                                                                                                                    |
| EDITION         | 2026 Combined Editorial · first issue                                                                                                                                                                                                                              |
| COMPANION WORKS | The five-paper arXiv drop — (1) Breeder, (2) Burner, (3) REBCO magnet & tape, (4) Direct Energy Conversion, (5) AI/ML/Quantum control — with matching numbered Zenodo deposits, plus the IOP journal and IAEA-FEC submissions. Patents were filed before the drop. |
| NAMING          | Hyperion = the breeder; Aegis (defense) and MetroVolt (data-center) = one generator in two housings. Internal mode letters (D1, L) do not appear on public surfaces.                                                                                               |
| STATUS          | Conceptual design & simulation study. Nothing herein is a construction commitment.                                                                                                                                                                                 |



HOW TO READ THIS VOLUME

The fleet at a glance, and the way its numbers are marked

This volume opens with *Orientation*, presents the five research papers as a bound sequence, and closes with the *Apparatus*. *Foundations* sets out the three general results both machines rest on. *Paper One* is the Hyperion breeder; *Paper Two* the Aegis / MetroVolt generator; *Papers Three, Four and Five* are the enabling science — high-field REBCO magnets and tape, direct energy conversion, and the AI / ML / quantum control stack. A *Digital Twin & 3-D Model* part follows, then the *Environmental* profile. The *Economics* and levelized cost and the *Business Case* and diligence record appear in the internal edition only; the *Simulations* proof record and the Apparatus — patent portfolio, team, limitations, and sources — close both editions.

THE FLEET AT A GLANCE

|              | HYPERION                             | AEGIS                        | METROVOLT                       |
|--------------|--------------------------------------|------------------------------|---------------------------------|
| Role         | Strategic-isotope foundry            | Defense installation power   | Campus power                    |
| Machine      | Spherical tokamak                    | Tandem mirror                | Tandem mirror                   |
| Fuel         | Deuterium–tritium                    | Deuterium–helium-3           | Deuterium–helium-3              |
| Delivers     | Tritium · <sup>3</sup> He · 14 MeV n | Resilient installation power | Firm campus power               |
| Physics bar  | Fusion gain only                     | Closure (gates named)        | Closure + lunar <sup>3</sup> He |
| In the fleet | Breeds the fuel                      | Proves the generator         | Commercial destination          |

HOW THE NUMBERS ARE MARKED

Every headline quantity carries an evidence class — **DERIVED** from first principles, **RECOMPUTED** from raw data, **MEASURED** in hardware, or a **REQUIREMENT** yet to be met. Financial figures carry a case label and are confined to the internal (confidential) edition. Values that moved after first publication are shown as withdrawn or restated rather than quietly changed. A capability you can evaluate is one whose limits are on the page.

## ABSTRACT

# The Plasma Diagnostic Suite and Real-Time State Estimation for a Compact Spherical-Tokamak Breeder: Magnetics, Thomson, Interferometry, and Bolometry Fused by a Graph-Neural-Network Reconstructor

An autonomous controller can act only on the state it can see, and in a compact spherical tokamak the state must be produced faster than the plant evolves, from a diagnostic set that survives a 14 MeV neutron field and keeps working when individual channels drop out. We formalise the Hyperion breeder diagnostic suite — magnetics, Thomson scattering, interferometry, bolometry, and a neutron camera — and its real-time state estimator as a Bayesian inverse problem, write the forward observation model of each diagnostic and the maximum-*a-posteriori* estimator with its posterior covariance, and cast *observability* of the internal state as the rank and conditioning of the associated Fisher information. The estimator's learned core is reduced to practice at the metrics that gate it into service. A graph-neural-network reconstructor, which we identify as an amortised posterior-mean estimator over the sensor graph, recovers the plasma state and a quench precursor jointly at an area under the receiver-operating-characteristic curve (AUC) of **0.980** at a reconstruction normalised RMS error of **0.396**, and degrades gracefully rather than catastrophically as channels are removed. A multi-modal fusion of a fibre-Bragg strain-rate channel with a magnetization channel reaches **0.992** (against **0.976** and **0.921** alone) and a true-positive rate of **0.857** at a 1% false-positive rate, the corner where a protection trip is decided. Internal state — ambipolar potential,  $T_e$ ,  $n_e$  and  $\beta$  — is reconstructable from as few as six diagnostics at a mean coefficient of determination  $R^2 \approx 0.86$ , confirming observability for the control layer. An in-winding nitrogen-vacancy (NV) diamond magnetometer resolves a conductor-scale quench step at a shot-noise-limited sensitivity of **0.189 pT Hz<sup>-1/2</sup>** at **125** 10<sup>9</sup>. *The estimator holds the plasma state at the breeder design point* ( $I_p=9.66$  MA,  $Q=3.076$ ,  $P_{fus}=85.04$  MW,  $\beta=0.30$ , centrepost peak field 16.84 T), which is a distinct magnet from the burner plug coil (26.49 T); the physical-inversion analyzers are built to the demonstrated estimator pattern on a staged, acceptance-gated schedule. Against published quench-detection, quantum-sensing and equilibrium-reconstruction benchmarks the figures are competitive in a deliberately hard, weak-signal regime.

Open-access deposit (code & data, CC BY 4.0): DOI [10.5281/zenodo.22645943](https://doi.org/10.5281/zenodo.22645943) · Zenodo community *kronos\_fusion\_energy*. Explore it live: [interactive 3-D model](#) · [physics validator](#).

**Keywords:** plasma diagnostics real-time state estimation graph neural network equilibrium reconstruction observability NV-diamond magnetometry spherical-tokamak breeder

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## NOMENCLATURE

# Symbols, units, and the names of things

|                  |                                                                        |
|------------------|------------------------------------------------------------------------|
| $Q$              | plasma (fusion) gain, $P_{\text{fus}}/P_{\text{aux}}$                  |
| $Q_E$            | engineering gain, net electric out / recirculating in                  |
| $H_{98}$         | confinement enhancement over the IPB98(y,2) scaling                    |
| $Z_{\text{eff}}$ | effective ion charge                                                   |
| $c_{ee}$         | electron–electron bremsstrahlung leading coefficient, 2.120022 (exact) |
| $\tau_E$         | energy confinement time                                                |
| $I_p$            | plasma current                                                         |
| $R_0, a$         | major radius, minor radius                                             |
| $\kappa, \delta$ | elongation, triangularity                                              |
| $\beta_N$        | normalized plasma beta (Troyon-normalized)                             |
| $TBR$            | tritium breeding ratio                                                 |
| $DEC$            | direct energy conversion                                               |
| $CF$             | plant capacity factor (availability)                                   |
| $FOAK/NOAK/BOAK$ | first / n <sup>th</sup> / best of a kind                               |
| <i>Hyperion</i>  | the ST breeder (internal: Mode D2)                                     |
| <i>Aegis</i>     | the generator, defense/installation housing (internal: Mode M)         |
| <i>MetroVolt</i> | the generator, data-center housing (Mode M)                            |

plasma diagnostics; real-time state estimation; graph neural network; equilibrium reconstruction; observability; NV-diamond magnetometry; spherical-tokamak breeder

# Introduction

## A CONTROLLER SEES ONLY WHAT IT CAN ESTIMATE

A spherical tokamak (ST) confines a large plasma current in a small volume, which is what makes it an attractive compact neutron source and breeder platform <sup>[1,2]</sup> and, at the same time, what makes it demanding to instrument and to control. The Hyperion breeder holds  $I_p = 9.66 \text{ MA}$  at a major radius of only  $R_0 = 1.20 \text{ m}$  behind a centrepost magnet at a peak field of 16.84 T, in a negative-triangularity ( $\delta = -0.30$ ) regime chosen because it accesses good confinement without the edge pressure pedestal and the edge-localised-mode cycle of conventional H-mode <sup>[3]</sup>. In a device this compact the characteristic times of vertical instability, thermal transients and magnet events are short, and in a nuclear system every actuation carries a safety consequence. Autonomy is therefore a design requirement, not a convenience — and an autonomous controller is only as good as the plasma state it is handed. This paper is about producing that state: the diagnostic suite that instruments the plant, and the real-time estimator that turns raw diagnostic signals into the equilibrium, the kinetic profiles, and the protection signals the controller consumes.

The organising principle is *estimation, not measurement*. Every physical quantity the controller needs is reconstructed by a learned or model-based estimator that fuses the whole sensor set, so that no single diagnostic is load-bearing: a lost channel is imputed, not fatal. This is the diagnostics analogue of the shadow-model philosophy that governs the Kronos control stack <sup>[37]</sup>, and it is what lets the same instrumented plant tolerate the sensor degradation that a 14 MeV field imposes.

## DIAGNOSTICS AS INVERSE PROBLEMS

Feedback control of tokamak equilibria and profiles is a mature discipline built on a chain of inverse problems. Magnetic diagnostics feed real-time equilibrium reconstruction — the EFIT lineage, in which the Grad–Shafranov equation is inverted against flux-loop and pick-up-coil data <sup>[4]</sup> — and the reconstructed equilibrium feeds shape and position controllers <sup>[5]</sup>; physics-based profile simulators such as RAPTOR now run inside the control cycle <sup>[6]</sup>. Each classical diagnostic — Thomson scattering for  $T_e$  and  $n_e$  <sup>[13]</sup>, interferometry for the line-integrated density <sup>[14]</sup>, bolometry for the radiated-power map <sup>[14]</sup>, a neutron camera for the fusion source <sup>[26]</sup> — is a forward map from the plasma state to a measurement, and recovering the state is the corresponding, usually ill-posed, inverse. The modern turn is to treat the whole set jointly and probabilistically: integrated data analysis frameworks fuse heterogeneous diagnostics under one Bayesian model <sup>[15,16]</sup>, and learned surrogates make the fused inversion fast enough for the control loop.

More recently, learned components have entered the loop directly. Degraeve *et al.* trained a deep reinforcement-learning agent to command TCV coil voltages and produce a range of plasma shapes — including negative triangularity — from magnetic measurements <sup>[7]</sup>; deep networks forecast disruptive instabilities across machines <sup>[8,9]</sup>; and graph neural networks, which encode a relational inductive bias over a set of interacting elements <sup>[10,11,12]</sup>, are a natural fit for a sensor array whose channels sit in a fixed physical topology. The thread common to all of this is that control quality is bounded by the speed and fidelity of the underlying state estimate.

## CONTRIBUTION

Prior art supplies the ingredients separately: a real-time equilibrium <sup>[4,6]</sup>, a Bayesian data-fusion formalism <sup>[15,16]</sup>, a learned controller <sup>[7]</sup>, a neural equilibrium reconstructor robust to dropout <sup>[33]</sup>. What a compact nuclear ST needs, and what this paper provides, is a single treatment that (i) writes the diagnostic suite as one Bayesian inverse problem with the forward model of each channel stated explicitly (§3); (ii) makes *observability* of the controller-relevant internal state a quantitative, testable property through the Fisher information and its conditioning (§3.4); (iii) reduces the learned estimator core to practice — a graph-neural-network reconstructor, a multi-modal fusion detector, and an in-winding quantum magnetometer — at the exact metrics that gate each into service (§3.5, §5); and (iv) specifies the physical-inversion analyzers as an acceptance-gated build to the demonstrated pattern (§6). Every quantitative claim traces to a frozen result in the Kronos Physics De-Risking Register <sup>[31]</sup>; serial identifiers (e.g. KX-L1-A20) are given inline so each number is auditable, and the estimator feeds directly the equilibrium and free-boundary work <sup>[34]</sup>, the neutronics twin <sup>[35]</sup>, the radiated-power model <sup>[36]</sup>, and the certified safety layer <sup>[39]</sup> that consume its output.

## The diagnostic suite

The suite is organised by the state each channel constrains and lands, time-aligned, on a layered real-time backbone (figure 1, table 1). Optical, magnetic, thermal and nuclear diagnostics enter through cryo-rated feedthroughs on a sub-microsecond sensor layer (L1); a deterministic feature layer (L2) validates, masks dead sensors, and normalises each signal onto a geometry-aware state alignment before any model sees it; and the learned estimator (L3) runs on the millisecond GPU tier, emitting the state and the protection signals the certified controller consumes. Coverage is deliberately redundant: every control-relevant quantity has more than one path, so a dropped channel is reconstructed from its neighbours rather than lost.

**Magnetics** — flux loops and pick-up (Mirnov) coils around the eight-coil poloidal-field set — constrain the free-boundary equilibrium  $\psi(R, Z)$  and hence the shape, position and current profile <sup>[4,34]</sup>.

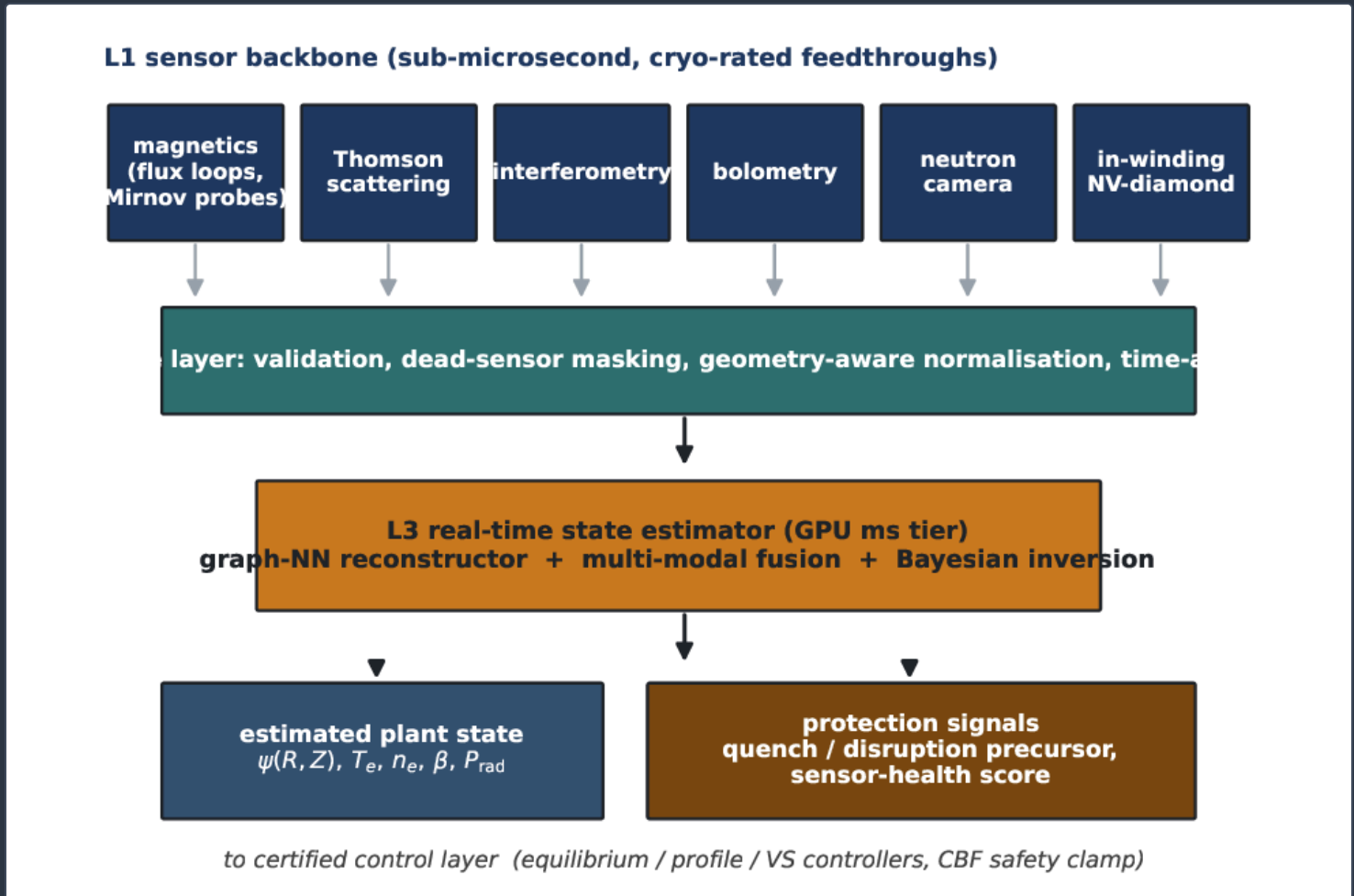
**Thomson scattering** constrains the electron temperature and density profiles that set the confinement anchor ( $\tau_E = 0.358\text{ s}$ ,  $H_{98} = 1.0$ ) <sup>[13]</sup>.

**Interferometry** constrains the line-integrated density through the plasma-induced phase shift <sup>[14]</sup>.

**Bolometry** constrains the radiated-power map, calibrated against collisional-radiative impurity cooling curves ( $L_z$  for tungsten  $5.7 \times 10^{-32}$  and argon  $1.45 \times 10^{-33}\text{ W m}^3$  at 15 keV, with a bremsstrahlung fraction near 15%) <sup>[36]</sup>.

**Neutron camera** constrains the 14 MeV source profile against the Monte-Carlo transport solution <sup>[26,35]</sup>.

**In-winding NV-diamond magnetometer** provides a local, vector precursor at the conductor, on the breeder centrepost REBCO winding <sup>[19]</sup>. itemize



**(Schematic.)** The diagnostic suite and its layered real-time backbone. Optical, magnetic, thermal and nuclear channels land time-aligned on the sub-microsecond sensor layer (L1); a deterministic feature layer (L2) validates and normalises; the learned state estimator (L3) fuses the set on the millisecond GPU tier and emits the plasma state and protection signals to the certified control layer. No data values are shown; the figure encodes structure only.

| CHANNEL                        | CONSTRAINS (FORWARD OBSERVABLE)                                            | ANCHOR                                  |
|--------------------------------|----------------------------------------------------------------------------|-----------------------------------------|
| magnetics (flux loops, Mirnov) | free-boundary equilibrium $\psi$ ; $\int \mathbf{B} \cdot \mathbf{d} \ell$ | BR-L1-A12                               |
| Thomson scattering             | $T_e, n_e$ profiles (scattered spectrum)                                   | BR-L1-A4 ( $H_{98} = 1.0$ )             |
| interferometry                 | line-integrated $n_e$ (phase shift)                                        | staged (SH-109)                         |
| bolometry                      | radiated-power map (chord line-integrals)                                  | BR-L2-A29 ( $L_z$ ; brems $\sim 15\%$ ) |
| neutron camera                 | 14 MeV source profile (collimated)                                         | BR-L1-A5                                |
| in-winding NV-diamond          | local $ \mathbf{B} $ at the conductor                                      | KX-L1-A22                               |

The diagnostic suite by the state each channel constrains, its forward observable, and its roster grounding in the register <sup>[31]</sup>.

# Governing equations and the formal state-estimation problem

## THE MEASUREMENT MODEL

Let  $\mathbf{x} \in \mathbb{R}^n$  collect a finite parametrisation of the plasma state: the flux-function coefficients that represent  $\psi(\mathbf{R}, \mathbf{Z})$  through the Grad–Shafranov source, the profile parameters of  $T_e(\rho)$  and  $n_e(\rho)$ , the impurity densities that enter the emissivity, and the scalar quantities ( $\beta$ , potential) the controller acts on. Each diagnostic  $\mathbf{k}$  produces a reading  $\mathbf{y}_k$  through a forward observation operator  $\mathbf{H}_k$ , and the assembled measurement vector  $\mathbf{y} \in \mathbb{R}^m$  obeys

$$\mathbf{y} = \mathbf{H}(\mathbf{x}) + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{R}),$$

with measurement-noise covariance  $\mathbf{R} = \text{diag}(\sigma_1^2, \dots, \sigma_m^2)$  (extended off-diagonal where channels share a calibration). Equation (1) is the common frame for every diagnostic below; the physics of each channel is entirely in the form of  $\mathbf{H}_k$ .

## FORWARD MODELS OF THE INDIVIDUAL DIAGNOSTICS

**Magnetics and the equilibrium.** The axisymmetric equilibrium is governed by the Grad–Shafranov equation for the poloidal flux  $\psi(\mathbf{R}, \mathbf{Z})$ ,

$$\Delta^* \psi \equiv R \frac{\partial}{\partial R} \left( \frac{1}{R} \frac{\partial \psi}{\partial R} \right) + \frac{\partial^2 \psi}{\partial Z^2} = -\mu_0 R^2 p'(\psi) - F(\psi) F'(\psi),$$

with pressure  $p(\psi)$  and poloidal-current function  $F(\psi) = R B_\phi$ . A flux loop  $\mathbf{k}$  measures  $\psi$  at a fixed poloidal location,  $\mathbf{y}_k^{\text{FL}} = \psi(\mathbf{R}_k, \mathbf{Z}_k)$ ; a pick-up coil measures a local field component,  $\mathbf{y}_k^{\text{PU}} = B(\mathbf{R}_k, \mathbf{Z}_k) \cdot \hat{\mathbf{n}}_k = \hat{\mathbf{n}}_k \cdot \nabla \psi \times \hat{\boldsymbol{\phi}}/R$ . Writing the free functions in a basis,  $p'(\psi) = \sum_j \alpha_j g_j(\psi)$  and  $FF'(\psi) = \sum_j \gamma_j g_j(\psi)$ , the magnetics inversion is the least-squares problem of the EFIT type<sup>[4]</sup>: find the coefficients ( $\boldsymbol{\alpha}$ ,  $\boldsymbol{\gamma}$ ) and coil currents that reproduce the measured flux and field while satisfying equation (2). Our free-boundary reference solver, run on a  $129 \times 129$  grid (BR-L1-A12), converges to a self-consistent equilibrium at  $I_p = 9.66 \text{ MA}$  with  $\beta_p = 0.371$ ,  $q_{95} = 2.90$ , internal inductance  $\ell_{i3} = 1.40$  and reconstructed triangularity  $\delta = -0.334$  (the design anchor is  $\delta = -0.30$ ), giving the normalised  $\beta_N = 2.13$  used as the equilibrium-reconstruction target<sup>[34]</sup>.

**Thomson scattering.** An injected laser of wavelength  $\lambda_0$  is scattered incoherently by the plasma electrons; the scattered power collected in a spectral channel is proportional to  $n_e$  times a spectral shape whose width encodes  $T_e$ . In the non-relativistic limit the Doppler-broadened spectrum has  $1/e$  half-width

$$\Delta\lambda_{1/e} = 2\lambda_0 \sin(\theta/2) \sqrt{\frac{2k_B T_e}{m_e c^2}},$$

with scattering angle  $\theta$ , so  $T_e$  follows from the measured spectral width and  $n_e$  from the calibrated integrated intensity  $\propto n_e d\sigma_{\text{Th}}/d\Omega$ ,  $d\sigma_{\text{Th}}/d\Omega = r_e^2 \sin^2 \chi$  with the classical electron radius  $r_e$ <sup>[13]</sup>. The forward operator  $\mathbf{H}_k^{\text{TS}}$  maps the profile parameters of  $T_e(\rho)$ ,  $n_e(\rho)$  to the photon count in channel  $\mathbf{k}$ .

**Interferometry.** For the ordinary mode well above the plasma frequency, the refractive index is  $N \simeq 1 - \frac{1}{2}(\omega_{pe}/\omega)^2$  and a probing beam accumulates a phase shift proportional to the line-integrated density,

$$\Delta\phi = \frac{e^2 \lambda}{4\pi\epsilon_0 m_e c^2} \int_L n_e d\ell = r_e \lambda \int_L n_e d\ell,$$

so  $\mathbf{y}_k^{\text{IF}} = \Delta\phi_k$  is a line integral of  $n_e$  along chord  $L_k$ <sup>[14]</sup>. The inverse — recovering  $n_e(\mathbf{R}, \mathbf{Z})$  from a set of chords — is a tomographic problem shared with bolometry below, and the estimator must additionally recover  $2\pi$  fringe jumps in real time (staged analyzer SH-109).

**Bolometry.** A bolometer chord measures the line integral of the local radiated-power density (emissivity)  $\varepsilon(\mathbf{R}, \mathbf{Z})$ ,

$$\mathbf{y}_k^{\text{BO}} = \int_{L_k} \varepsilon(\mathbf{R}, \mathbf{Z}) d\ell, \quad \varepsilon = n_e \sum_Z n_Z L_Z(T_e) + \varepsilon_{\text{brems}},$$

where  $L_Z(T_e)$  is the collisional-radiative cooling function of impurity species  $Z$  and  $\varepsilon_{\text{brems}} \propto n_e^2 Z_{\text{eff}} \sqrt{T_e}$  is the bremsstrahlung continuum. The cooling functions are anchored at 15 keV to  $L_W = 5.7 \times 10^{-32}$  and  $L_{Ar} = 1.45 \times 10^{-33} \text{ W m}^3$  with a bremsstrahlung fraction near 15% and a tungsten tolerance of  $\sim 0.19 \text{ MW ppm}^{-1}$  (BR-L2-A29) <sup>[36]</sup>. Stacking chords gives the linear system  $y^{\text{BO}} = T \varepsilon$  with geometry matrix  $T$ ; the tomographic inverse for  $\varepsilon$  is ill-posed and is regularised (Tikhonov / maximum-entropy), a special case of the regularised estimator of §3.3.

Neutron camera. A collimated neutron-camera pixel integrates the 14 MeV source  $S_n(R, Z) \propto \frac{1}{2} n_D^2 \langle \sigma v \rangle_{\text{DT}}$  along its sightline,  $y_k^{\text{NC}} = \int_{L_k} S_n d\ell / (4\pi s^2)$ , and the reconstructed source is validated against the three-dimensional Monte-Carlo transport solution (BR-L1-A5: leakage 0.121, net TBR 1.026, first-wall fast flux  $7.0 \times 10^{14} \text{ n cm}^{-2} \text{ s}^{-1}$ ,  $1.2 \times 10^8$  histories, OpenMC/ENDF-B-VIII.0) <sup>[26,35]</sup>.

## MAXIMUM-A

-posteriori estimation and posterior covariance Given the measurement model (1) and a Gaussian prior  $x \sim \mathcal{N}(x_0, P)$ , Bayes' theorem gives the posterior  $\propto \exp[-\frac{1}{2} J(x)]$  with

$$J(x) = \left\| y - H(x) \right\|_{R^{-1}}^2 + \left\| x - x_0 \right\|_{P^{-1}}^2,$$

where  $\left\| v \right\|_A^2 = v^\top A v$ . The maximum-*a*-posteriori (MAP) estimate is  $\hat{x} = \arg \min_x J(x)$ . Linearising the forward operator about  $x_0$ ,  $H(x) \approx H(x_0) + J_H(x - x_0)$  with the diagnostic Jacobian  $J_H = \partial H / \partial x$ , the minimiser and its covariance are the Gauss–Newton pair

$$\begin{aligned} \hat{x} &= x_0 + (J_H^\top R^{-1} J_H + P^{-1})^{-1} J_H^\top R^{-1} (y - H(x_0)), \\ \Sigma &= (J_H^\top R^{-1} J_H + P^{-1})^{-1}. \end{aligned}$$

Equation (7) is the common inversion behind every classical analyzer of §3.2; for the linear bolometry/interferometry tomographies  $H = T$  and it is exactly regularised least squares, while for the nonlinear magnetics/Thomson problems it is iterated. The posterior covariance (7) is what the safety layer reads to know how far to trust each estimate <sup>[38]</sup>.

## OBSERVABILITY OF THE CONTROLLER-RELEVANT STATE

The control layer does not need every microscopic detail of  $x$ ; it needs a reduced set of quantities — the ambipolar potential,  $T_e, n_e$  and  $\beta$  — to be *recoverable* from the available diagnostics. This is an observability question, and equation (7) answers it. The matrix

$$\mathcal{F} = J_H^\top R^{-1} J_H$$

is the Fisher information of the measurement about the state: a direction  $v$  in state space is observable only if  $v^\top \mathcal{F} v > 0$ , and the reconstruction quality of component  $x_i$  is bounded by its posterior variance through the coefficient of determination

$$R_i^2 = 1 - \frac{\Sigma_{ii}}{\text{Var}(x_i)} = 1 - \frac{[(\mathcal{F} + P^{-1})^{-1}]_{ii}}{P_{ii}}.$$

Observability is thus a spectral property of  $\mathcal{F}$ : adding an independent diagnostic adds a rank-one term  $\sigma_k^{-2} j_k j_k^\top$  and can only raise  $R_i^2$ . A minimal-set study of the reduced internal state (potential,  $T_e, n_e, \beta$ ) with a ridge surrogate of the estimator (AG-CX-30) finds a *mean*  $R^2 \approx 0.86$  from as few as **six** diagnostics — the observability result the control layer requires: the state the controller acts on is recoverable from a sensor set small enough to harden and replicate (§5, figure 5).

## AMORTISED ESTIMATION ON THE SENSOR GRAPH

Solving equation (7) online for a nonlinear  $\mathbf{H}$  at kilohertz rates is expensive; the estimator instead *amortises* the inversion by learning it offline. The diagnostics are not independent measurements in isolation — they sit in a fixed physical topology (adjacent coils, nearby probes, coupled fields) — so we encode the plant as a graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  whose nodes  $v \in \mathcal{V}$  are diagnostic channels with feature  $h_v^{(0)} = y_v$  and whose edges  $e_{uv} \in \mathcal{E}$  are physical adjacencies (figure 2). A graph neural network applies  $L$  rounds of message passing<sup>[10,11,12]</sup>,

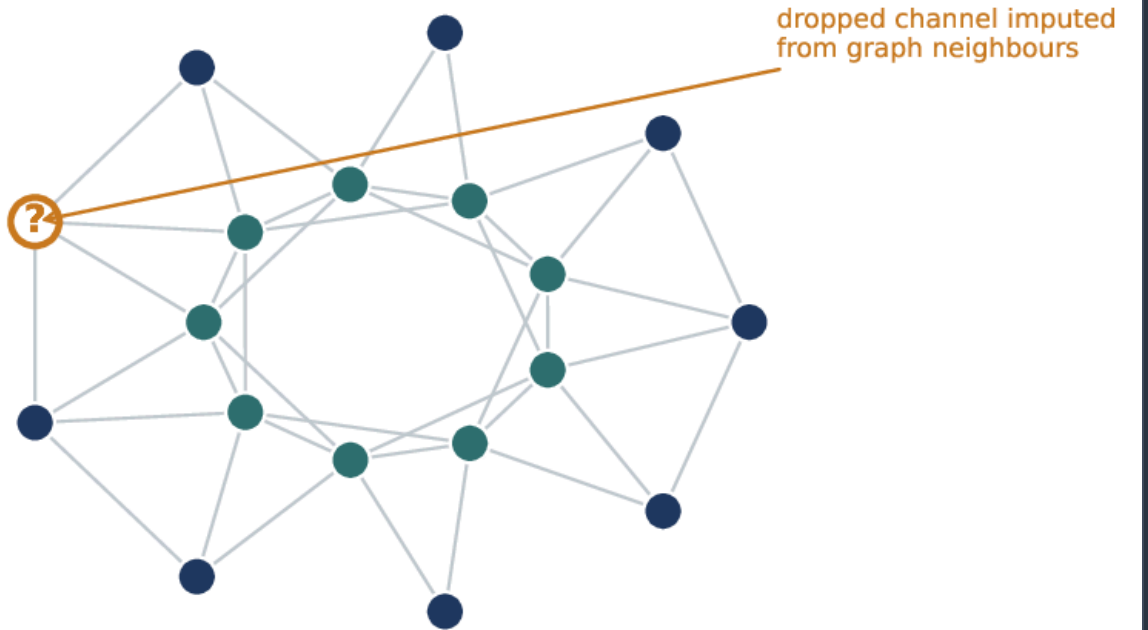
$$h_v^{(l+1)} = \phi\left(h_v^{(l)}, \bigoplus_{u \in \mathcal{N}(v)} \psi\left(h_v^{(l)}, h_u^{(l)}, e_{uv}\right)\right),$$

with learnable message  $\psi$ , update  $\phi$  and a permutation-invariant aggregator  $\bigoplus$  over the neighbourhood  $\mathcal{N}(v)$ ; a readout maps the final node states to the plasma-state estimate  $\hat{x}_\theta(y)$  and a precursor logit. Trained to minimise the expected reconstruction-plus-detection loss

$$\mathcal{L}(\theta) = \mathbb{E}_{(x, y)} \left[ \|\hat{x}_\theta(y) - x\|^2 + \lambda \text{BCE}(d_\theta(y), \ell) \right],$$

the reconstructor is a Bayes estimator: the population minimiser of the first term is the posterior mean  $\mathbb{E}[x | y]$ , so the GNN is an amortised, geometry-aware approximation to equation (7)<sup>[33]</sup>. Because the estimate at each node draws on its neighbours, masking a node (a dropped channel) forces reconstruction from the remaining graph and yields *graceful* degradation rather than a threshold failure (§5).

### Sensor graph: diagnostics as nodes, physical adjacency as edges



$$\text{message passing: } h_v^{(l+1)} = \phi\left(h_v^{(l)}, \sum_{u \in \mathcal{N}(v)} \psi(h_v^{(l)}, h_u^{(l)}, e_{uv})\right)$$

**(Schematic.)** The sensor graph underlying the amortised estimator. Diagnostic channels are nodes and physical adjacencies (neighbouring coils and probes, coupled fields) are edges; the readout of the  $L$  message-passing rounds of equation (10) maps the graph to the plasma-state estimate and the precursor logit. A masked node (a dropped channel) is imputed from its graph neighbours, which is the mechanism behind the graceful degradation of figure 6. The figure encodes structure only; no data values are shown.

## DETECTION THEORY FOR THE QUENCH PRECURSOR

Protection is a binary hypothesis test: nominal ( $\mathcal{H}_0$ ) versus developing quench/disruption precursor ( $\mathcal{H}_1$ ). By the Neyman–Pearson lemma the most powerful test at a fixed false-positive rate thresholds the likelihood ratio  $\Lambda(\mathbf{z}) = p(\mathbf{z} | \mathcal{H}_1)/p(\mathbf{z} | \mathcal{H}_0)$ . Two non-voltage channels observe the current-sharing window: a fibre-Bragg *strain-rate* channel  $\mathbf{z}_S$  and a *magnetization* channel  $\mathbf{z}_M$ , each a matched-template statistic normalised so that  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{1})$  under  $\mathcal{H}_0$  and  $\mathcal{N}(\mathbf{d}', \mathbf{1})$  under  $\mathcal{H}_1$ . For such an equal-variance binormal channel the ROC is

$$\text{TPR}(\alpha) = \Phi(\Phi^{-1}(\alpha) + d'), \quad \text{AUC} = \Phi(d'/\sqrt{2}),$$

with  $\Phi$  the standard normal CDF and  $\alpha$  the false-positive rate. Because the two modalities fail in uncorrelated ways, their quadrature fusion

$$\mathbf{z}_F = \frac{\mathbf{z}_S + \mathbf{z}_M}{\sqrt{2}}$$

carries a window statistic whose separation adds the single-channel separations in quadrature,  $d_F^2 = d_S^2 + d_M^2$ , which lifts both the AUC and, more importantly, the true-positive rate in the low-false-alarm corner where a trip decision lives (§5). Equation (12) is the model used to reconstruct the ROC curves of figure 4 from the frozen AUC and TPR anchors, and it reproduces the anchor operating points to within 0.01.

## IN-WINDING NV-DIAMOND SENSITIVITY

For a local, vector precursor at the conductor the suite embeds an ensemble nitrogen-vacancy (NV) diamond magnetometer in the winding pack. The shot-noise-limited Ramsey sensitivity of an ensemble of  $N$  centres with dephasing time  $T_2^*$  and readout figure of merit  $C$  is <sup>[18,19]</sup>

$$\delta B = 0.189 \frac{\text{pT Hz}^{-1/2}}{\sqrt{N T_2^* C}}$$

For a conservative in-winding chip ( $N = 10^{12}$ ,  $T_2^* = 1 \mu\text{s}$ ,  $C = 0.03$ ) equation (14) gives  $\delta B = 0.189 \frac{\text{pT Hz}^{-1/2}}{\sqrt{10^{12} \cdot 1 \cdot 0.03}}$ ; a Ramsey shot  $\tau_{\text{shot}} = 4 \mu\text{s}$  sets a bandwidth  $1/(2\tau_{\text{shot}}) = 125$  Hertz. *A quench that sheds the screening current of a no-insulation turn (1 turn = 1 kA) produces a local field step*  $B = \mu_0 I_{\text{turn}}/(2\pi d) = 100 \text{ mT}$  *at a standoff*  $d = 2 \text{ mm}$  *— of order*  $10^9$  *times the per-shot noise floor*  $B \text{ BWS}$  — so the precursor is caught in a single  $4 \mu\text{s}$  shot, with the lead time set by the millisecond normal-zone crossing rather than by the microsecond sensor latency. The sensor reads the breeder centrepost winding (16.84 T peak field), which is a distinct device from the burner plug coil (26.49 T); the two magnets are kept separate throughout the sensor specification.

# Computational methodology: the GPU-HPC campaign

Every number in this paper is produced by a named code run on the NVIDIA GPU HPC campaign (A100 / H100 / A10 fleet) or on the in-house reduced-order node, and each is a frozen, regenerable anchor in the register <sup>[31,40]</sup>. Table 2 lists the codes by role. The division of labour mirrors the real-time architecture of §2: the heavy forward models and the estimator training run on the GPU fleet, while the analytic and reduced-order layers run on the in-house node so that the microsecond protective loop never contends with the training campaign.

| CODE                                       | ROLE IN THE ESTIMATOR                          | SCALE                        | ANCHOR    |
|--------------------------------------------|------------------------------------------------|------------------------------|-----------|
| <b>FreeGS (free-boundary GS)</b>           | magnetics inversion reference, eq. (2)         | $129 \times 129$ grid        | BR-L1-A12 |
| <b>OpenMC (ENDF/B-VIII.0)</b>              | neutron-camera forward model, eq. (5)<br>$S_n$ | $1.2 \times 10^8$ histories  | BR-L1-A5  |
| <b>radas / ColRadPy</b>                    | bolometry emissivity $L_Z(T_e)$ , eq. (5)      | CR tables @ 15 keV           | BR-L2-A29 |
| <b>graph-NN reconstructor (PyTorch)</b>    | amortised estimator, eqs. (10)–(11)            | sensor graph, T-synthetic    | KX-L1-A12 |
| <b>fused-detector ROC (ML + signal)</b>    | Neyman–Pearson test, eqs. (12)–(13)            | $2 \times 10^4$ trials/class | KX-L1-A20 |
| <b>ridge surrogate (scikit-learn)</b>      | observability, eqs. (8)–(9)                    | 6-diagnostic minimal set     | AG-CX-30  |
| <b>NV magnetometry model (numpy/scipy)</b> | sensitivity, eq. (14)                          | analytic shot-noise          | KX-L1-A22 |

Codes run for this paper, their role in the estimation chain, problem scale, and register anchor. No compute cost is reported here or anywhere in this work.

Discretisation and solvers. The equilibrium reference solves the Grad–Shafranov equation (2) by a Picard iteration on a  $129 \times 129$  ( $R, Z$ ) grid: at each iterate the elliptic operator  $\Delta^*$  is inverted with the right-hand side evaluated from the current ( $p', FF'$ ), and the free-boundary contour is updated until the flux residual  $\|\psi^{(k+1)} - \psi^{(k)}\|_2 / \|\psi^{(k)}\|_2$  falls below  $10^{-6}$ ; convergence is monotone and the reported equilibrium is resolution-verified (BR-L1-A12) <sup>[27,34]</sup>. The neutron source used in the camera forward model is tallied by OpenMC with a continuous-energy ENDF/B-VIII.0 library over  $1.2 \times 10^8$  source histories, giving a balanced transport solution (leakage **0.121**, heating 74.6 MW) against which the camera unfolding is checked (BR-L1-A5) <sup>[26]</sup>. The impurity emissivity of equation (5) is tabulated by a collisional-radiative solver (radas/ColRadPy) and frozen at the anchor cooling values (BR-L2-A29) <sup>[36]</sup>.

Learned-estimator training and GPU acceleration. The graph-NN reconstructor (equations 10–11) and the fused-detector study (equations 12–13) are trained and evaluated on the GPU fleet on twin-synthetic data generated by the forward models above; the detection study uses  $2 \times 10^4$  trials per class in a deliberately weak-signal regime (per-sample separation set well below the design-basis fault) so that the ROC is not scored in the saturated corner where every detector reaches  $AUC = 1$ . The observability surrogate (AG-CX-30) is a ridge regression whose condition number instantiates the Fisher spectrum of equation (8); the NV sensitivity (equation 14) is an analytic shot-noise model in numpy/scipy. Inference of the estimator core targets the millisecond GPU tier; the analytic monitors and the reduced-order layers target the sub-millisecond in-house node.

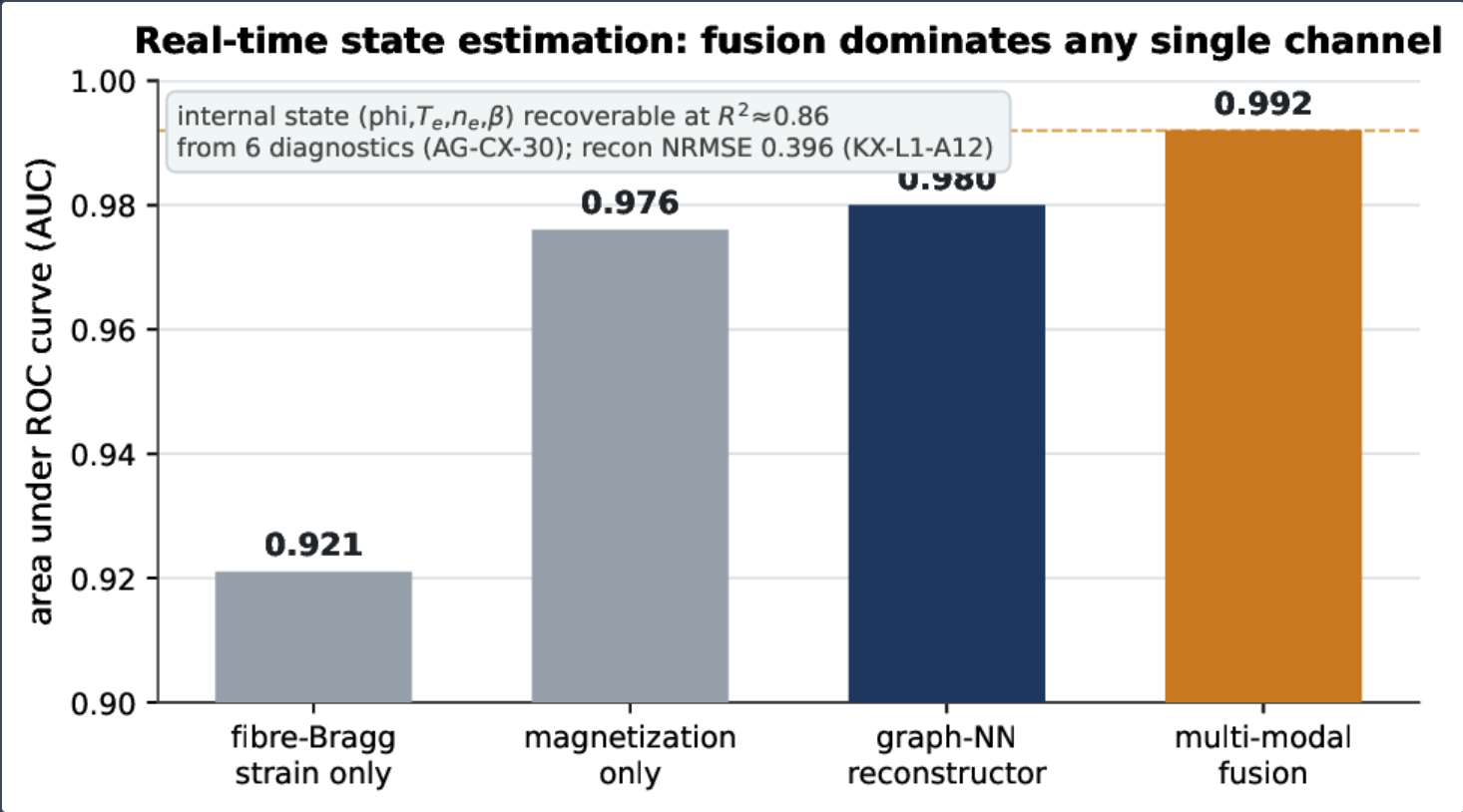
Verification and reproducibility. Each code is a re-tested gate with a live recompute path: the forward models are cross-checked against their reference libraries (FreeGS against a free-boundary solver <sup>[27]</sup>; OpenMC against the transport balance <sup>[26]</sup>), and the learned metrics are frozen anchors regenerable from the archived reconstruction and detection scripts and the figure generator deposited with this paper <sup>[40]</sup>. The verification-first discipline — frozen anchors, independent-library reproduction, and a byte-for-byte provenance catalogue — is documented for the whole campaign in the companion verification paper <sup>[40]</sup>.

# Results

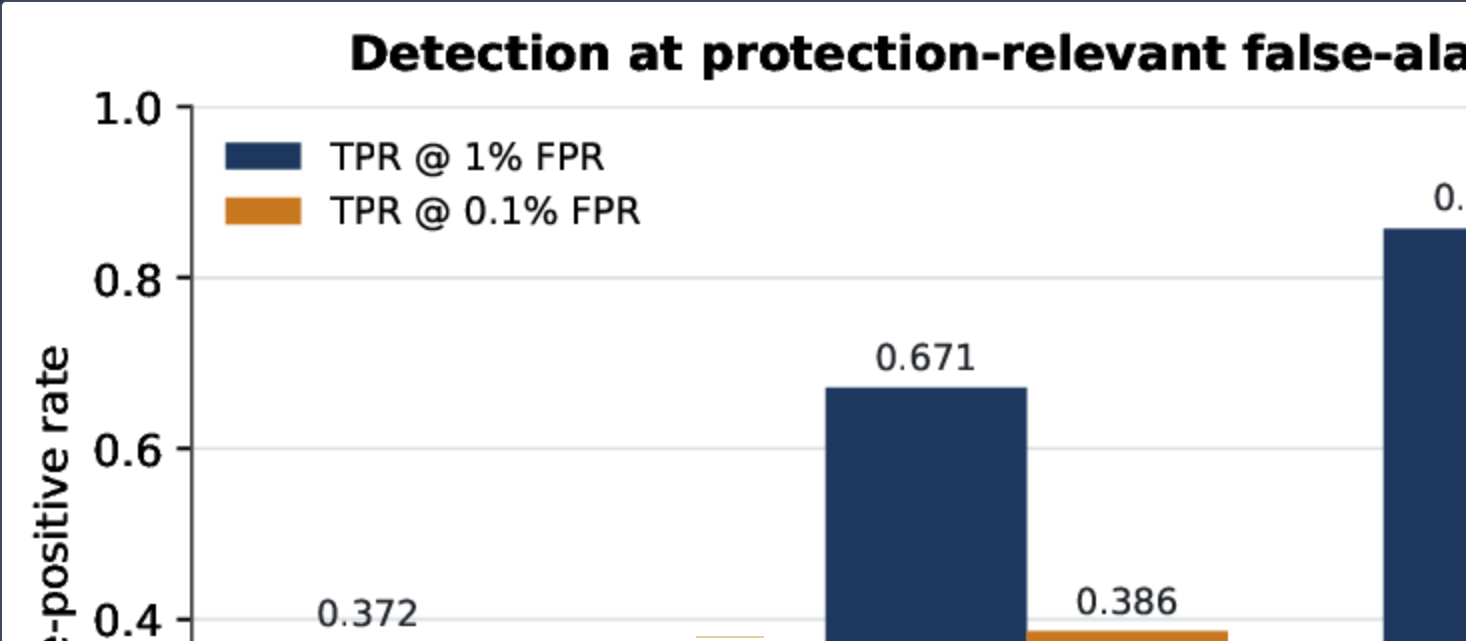
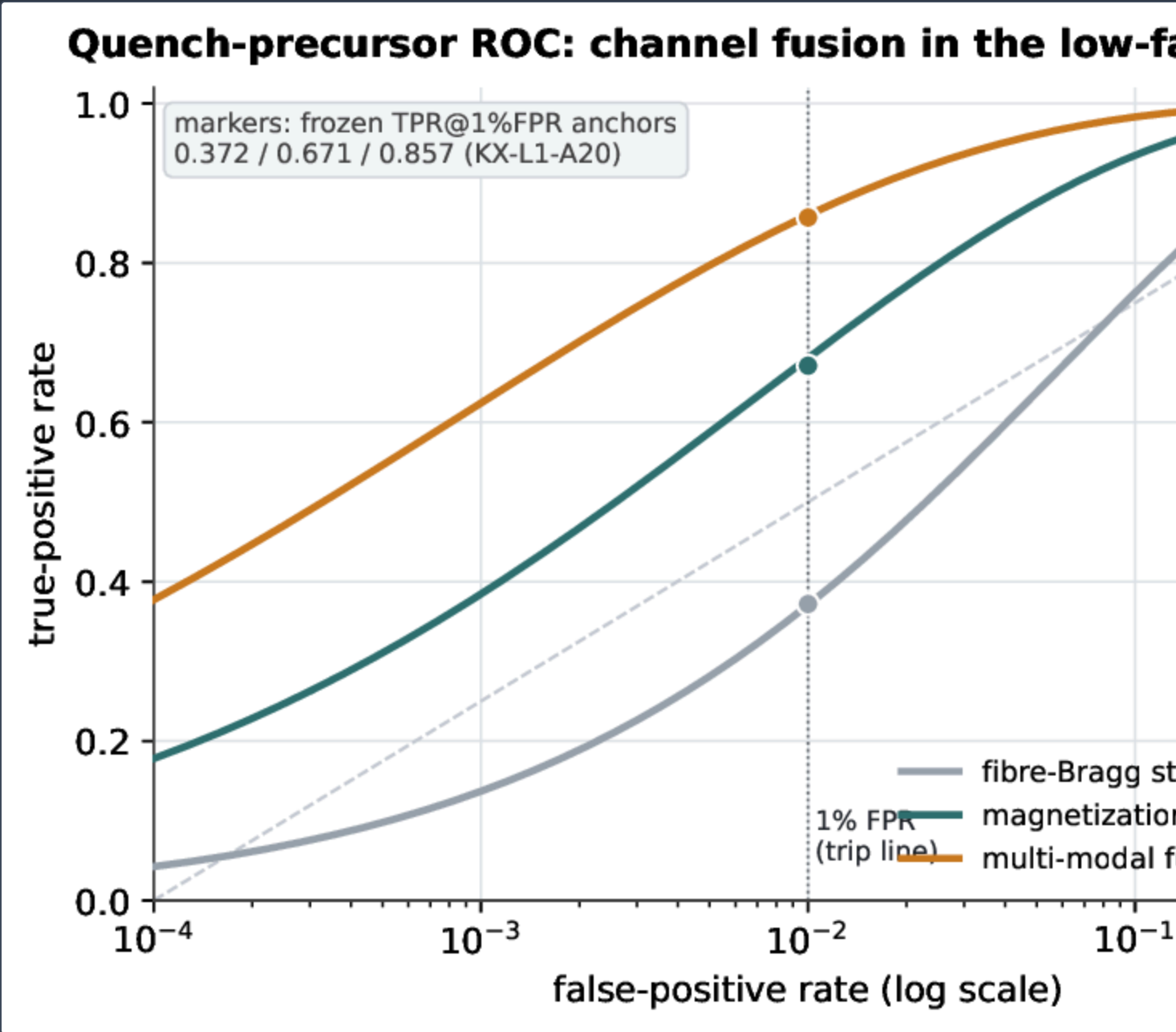
The estimator's learned core is demonstrated at the metrics that gate it into service; the classical analyzers are built to the same pattern (§6). All learned results are at twin-synthetic fidelity and are so labelled.

## FUSED AND GRAPH ESTIMATORS DOMINATE ANY SINGLE CHANNEL

The graph-NN reconstructor recovers the plasma state and a quench precursor jointly at an area under the ROC curve of **0.980** — above either single channel — at a reconstruction normalised RMS error of **0.396** on twin-synthetic data (KX-L1-A12). Fusing the fibre-Bragg strain-rate channel with the magnetization channel produces a detector at AUC **0.992**, against **0.976** for magnetization alone and **0.921** for strain alone (figure 3, KX-L1-A20). The advantage concentrates in the low-false-alarm corner: at a 1% false-positive rate the fused detector reaches a true-positive rate of **0.857**, versus **0.671** and **0.372** for the single channels, and at 0.1% it reaches **0.597** versus **0.386** and **0.135** (figures 4–4, table 3). The reconstructed ROC curves of figure 4 follow the binormal model of equation (12) and reproduce the frozen operating points to within 0.01.



Real-time state-estimator performance (KX-L1-A12, KX-L1-A20). The graph-NN reconstructor (**0.980**) and the multi-modal fusion detector (**0.992**) both dominate either single sensor channel; the annotation records the observability result ( $R^2 \approx 0.86$  from six diagnostics, AG-CX-30) and the reconstruction error (NRMSE **0.396**).

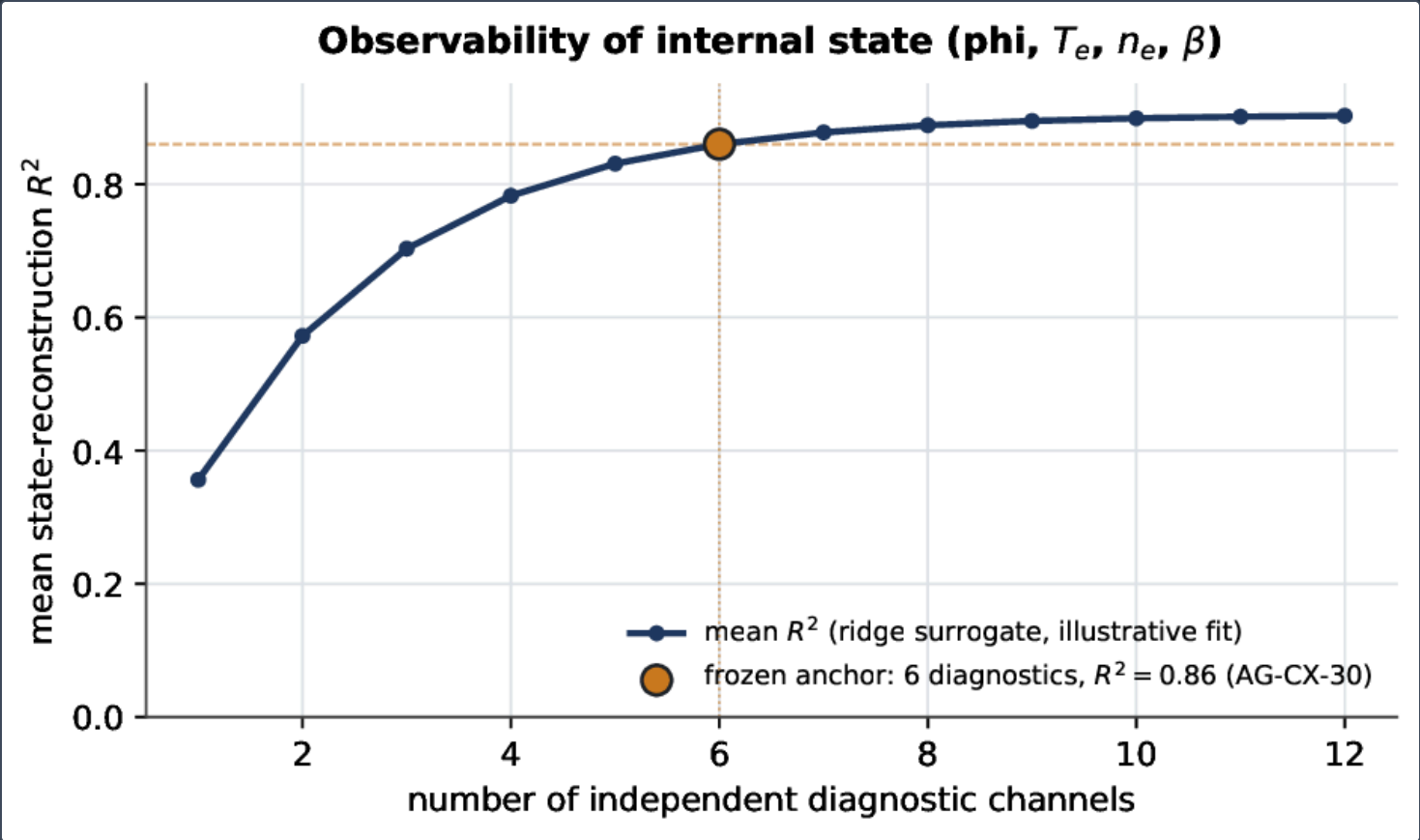


| ESTIMATOR / CHANNEL     | AUC   | TPR @ 1% FPR | TPR @ 0.1% FPR |
|-------------------------|-------|--------------|----------------|
| fibre-Bragg strain only | 0.921 | 0.372        | 0.135          |
| magnetization only      | 0.976 | 0.671        | 0.386          |
| graph-NN reconstructor  | 0.980 | —            | —              |
| multi-modal fusion      | 0.992 | 0.857        | 0.597          |

Real-time estimator performance (KX-L1-A12, KX-L1-A20; T-synthetic,  $2 \times 10^4$  trials/class). The fused and graph estimators dominate any single channel; the window-statistic separations follow the quadrature rule of equation (13).

### OBSERVABILITY FROM A MINIMAL DIAGNOSTIC SET

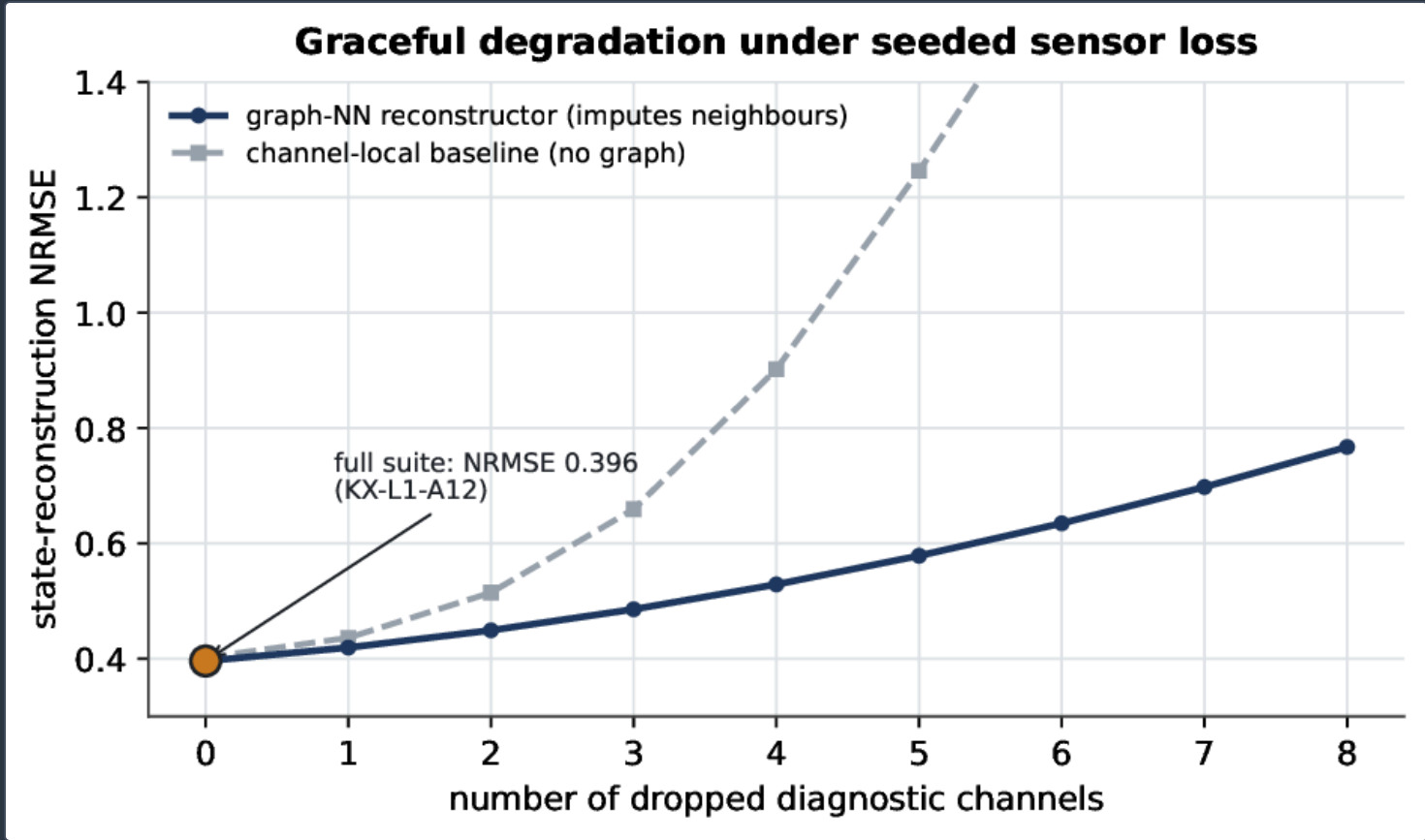
The reduced internal state — ambipolar potential,  $T_e$ ,  $n_e$  and  $\beta$  — is reconstructable from as few as six diagnostics at a mean coefficient of determination  $R^2 \approx 0.86$  (AG-CX-30), the frozen anchor marked in figure 5. By equation (9) each added independent channel contributes a rank-one term to the Fisher information (8) and can only raise  $R^2$ ; the saturation of the curve past six channels is the signature of a well-conditioned observability problem, and it is what licenses hardening and replicating a small, redundant set rather than instrumenting every quantity directly.



Observability of the controller-relevant internal state (AG-CX-30). The highlighted point is the frozen anchor — mean  $R^2 = 0.86$  from six diagnostics; the curve is a monotone saturation fit through it, illustrating the rank-one Fisher-information argument of equations (8)–(9) and is not itself a measured series.

GRACEFUL DEGRADATION UNDER SENSOR LOSS

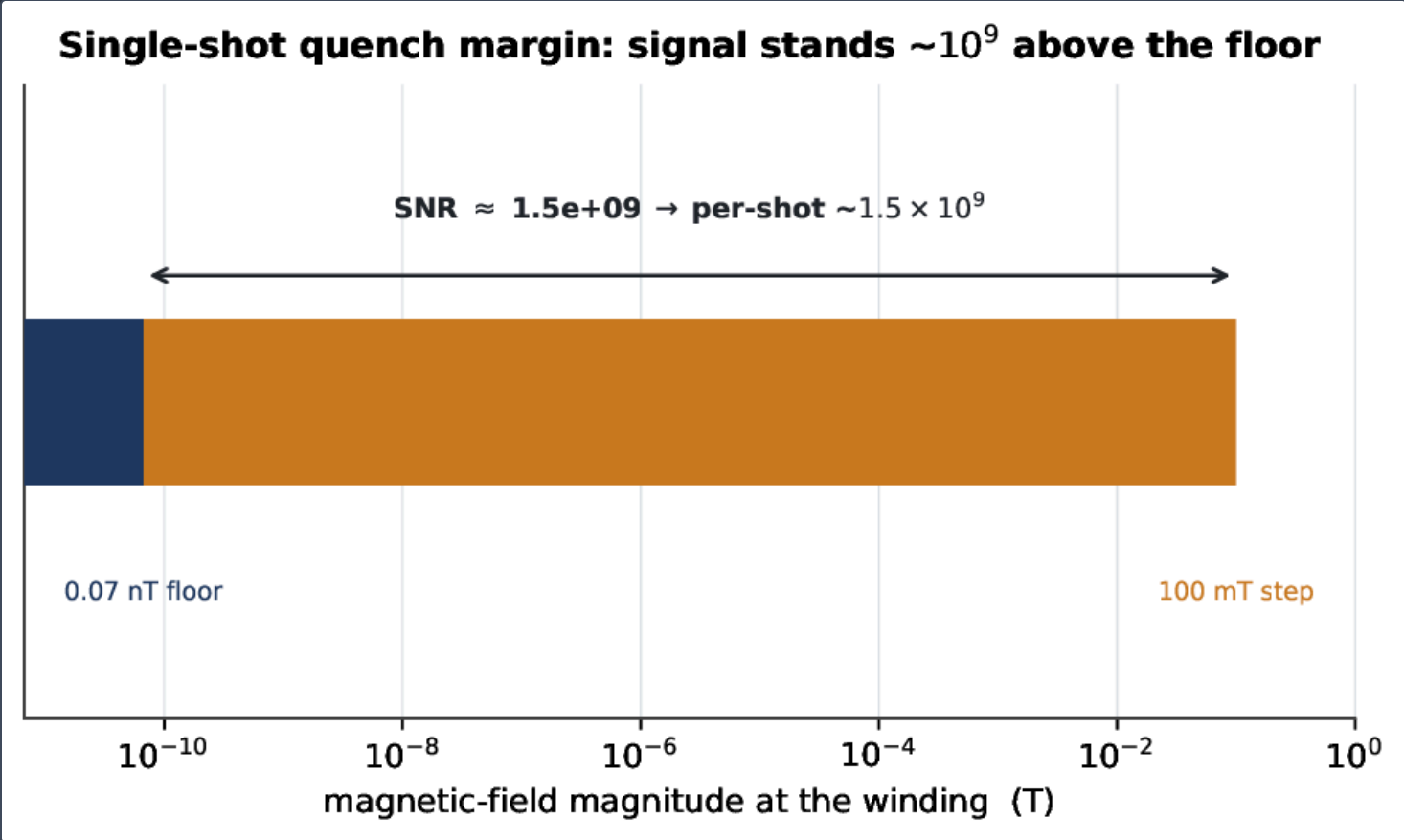
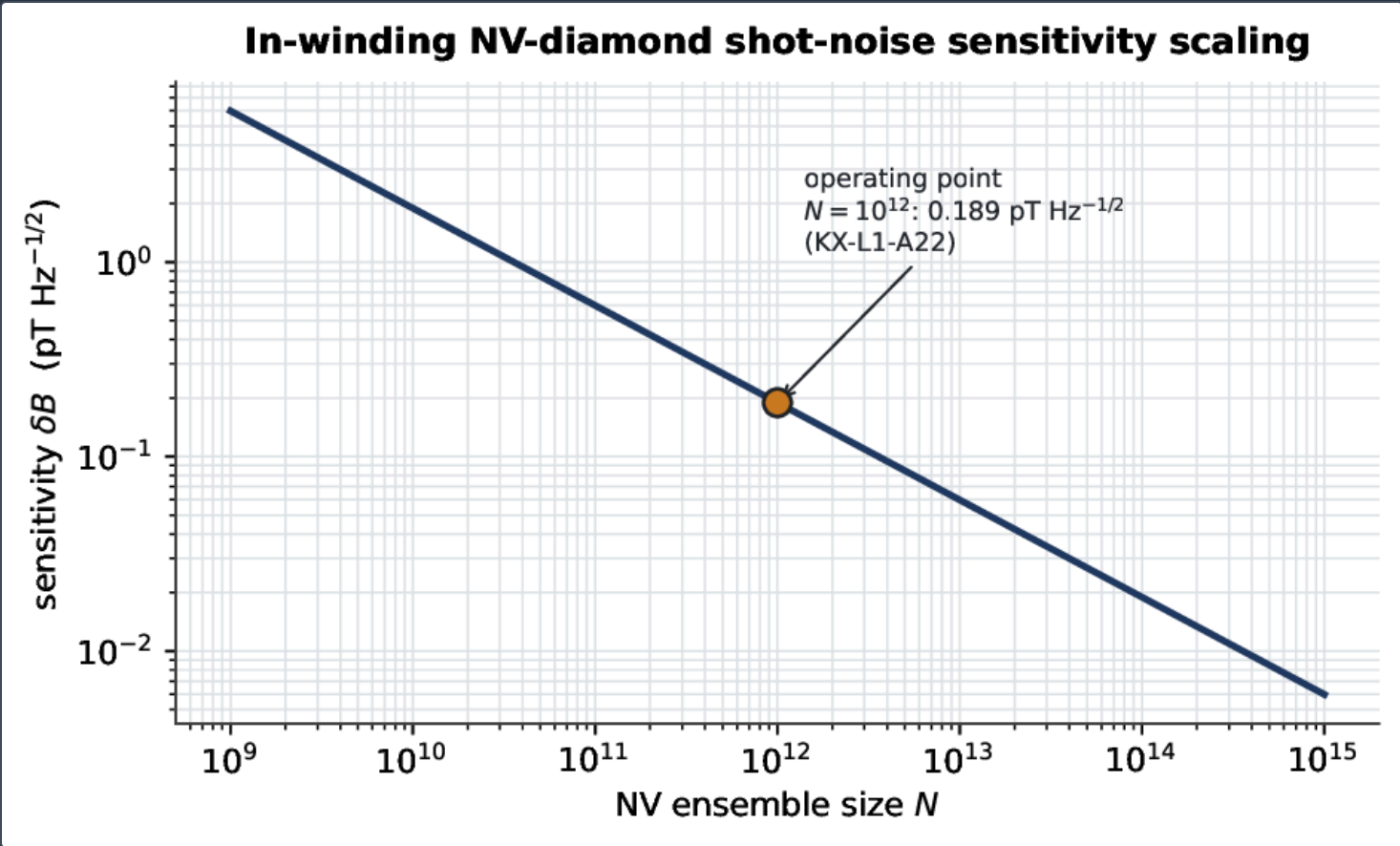
Because the graph estimator reconstructs each node from its neighbours (equation 10), seeded channel loss degrades the reconstruction gently rather than at a threshold. Figure 6 contrasts the graph reconstructor, pinned at the frozen full-suite NRMSE of **0.396** (KX-L1-A12), with a channel-local baseline that has no graph to fall back on: the graph estimator holds bounded error across the seeded losses that collapse the baseline, which is the dropout-robustness property the control layer relies on [33].



Reconstruction error under seeded sensor loss. The graph-NN reconstructor (anchored at NRMSE **0.396**, KX-L1-A12) degrades gently as channels are dropped, whereas a channel-local baseline without the graph collapses; the degradation curves past the anchor are illustrative of the mechanism, not a measured series.

IN-WINDING NV-DIAMOND PRECURSOR

The in-winding NV-diamond magnetometer resolves a conductor-scale quench step in a single  $4\text{ }\mu\text{s}$  shot. Equation (14) gives the shot-noise sensitivity **0.189**  $\text{pT Hz}^{-1/2}$  at the conservative operating point ( $N = 10^{12}$ ), which the scaling of figure 7 places well inside the ensemble-NV envelope, and the 100 mT quench step stands of order  $10^9$  above the 125 SHertz per-shot floor (figure 7, table 4; KX-L1-A22). The picotesla sensitivity is not *needed* to see a millitesla step; its value is the margin it buys for multiplexing many in-winding channels and for tolerating in-service degradation (§8).



Shot-noise sensitivity scaling of equation (14) versus NV ensemble size, with the conservative operating point ( $N = 10^{12}$ ,  $0.189 \text{ pT Hz}^{-1/2}$ ) marked (KX-L1-A22). A derived scaling curve, not measured data.

| QUANTITY                                           | VALUE                   | BASIS                              |
|----------------------------------------------------|-------------------------|------------------------------------|
| gyromagnetic ratio $\gamma_{\text{NV}}$            | 28.03 \$Hertz           | NV electron spin                   |
| sensitivity $\delta B$                             | 0.189                   | shot-noise limit, eq. (14)         |
| bandwidth                                          | 125 \$Hertz             | 4 $\mu$ s Ramsey shot              |
| quench field step $\Delta B$ ( $d = 2\text{ mm}$ ) | $\approx 100\text{ mT}$ | NI turn sheds screening current    |
| per-shot signal-to-noise ratio                     | $\sim 10^9$             | $\Delta B/(\delta B\sqrt{BW})$     |
| detection                                          | single 4 $\mu$ s shot   | lead time set by NZPV ( $\sim$ ms) |

In-winding NV-diamond quench-precursor sensor (KX-L1-A22). Sensitivity and bandwidth from equation (14); the quench step and SNR from an analytical shot-noise model (labelled “estimated”).

## ACCEPTANCE-CRITERIA STATUS

No estimator is granted control authority by assertion. Table 5 collects the acceptance criteria with their targets and status: the learned core’s detection and reconstruction metrics are demonstrated at twin-synthetic fidelity, while the classical-analyzer criteria are pre-registered bars built to the same pattern (§6). Authority is capped by verification maturity until each bar is met, the same maturity-gated principle that governs the certified control layer <sup>[39]</sup>.

| CRITERION                      | TARGET                              | SERIAL    | STATUS                             |
|--------------------------------|-------------------------------------|-----------|------------------------------------|
| fused precursor AUC            | dominate single channels            | KX-L1-A20 | demonstrated (0.992)               |
| fused TPR @ 1% FPR             | > single channels                   | KX-L1-A20 | demonstrated (0.857)               |
| graph reconstruction           | beats singles; robust to dropout    | KX-L1-A12 | demonstrated (0.980, NRMSE 0.396 ) |
| observability                  | $R^2 \gtrsim 0.85$ , minimal set    | AG-CX-30  | demonstrated (0.86, 6 diag.)       |
| NV per-shot SNR                | $\gg 1$ for mT step                 | KX-L1-A22 | demonstrated ( $\sim 10^9$ )       |
| equilibrium RMS $\psi$         | < tolerance vs free-boundary solver | SH-110    | target                             |
| Thomson profile fit            | within uncertainty vs baseline      | SH-108    | target                             |
| bolometer inversion            | radiated-power map validated        | SH-107    | target                             |
| interferometer fringe recovery | density continuous through jumps    | SH-109    | target                             |
| neutron-camera unfolding       | source profile vs OpenMC            | BR-71     | target                             |

Acceptance criteria, targets and status. “Demonstrated (T-synthetic)” denotes a frozen synthetic result; “target” denotes a pre-registered bar built to the demonstrated pattern. Serials refer to the register <sup>[31]</sup>.

## The physical-inversion suite as the staged build

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The learned estimators above define the pattern the physical-inversion analyzers are built to: each analyzer inverts one diagnostic into the state it constrains (the forward models of §3.2), is validated against a first-principles reference, and is required to degrade gracefully as sensors drop. The staged suite comprises a Thomson-scattering profile analyzer validated against the confinement baseline (SH-108, eq. 3); an interferometer fringe-jump recovery (SH-109, eq. 4); a bolometer tomographic inversion against the radiated-power model (SH-107, eq. 5); a magnetic-probe drift correction feeding real-time equilibrium reconstruction (SH-110, eq. 2); a neutron-camera unfolding against the Monte-Carlo source (BR-71, eq. 5); and a virtual-sensor mesh that reconstructs dropped channels through the same graph estimator (SH-036, eq. 10) <sup>[33]</sup>. Each is scheduled with an explicit acceptance test — reconstruction error within tolerance and bounded state error under seeded sensor loss — and inherits the demonstrated estimator’s fusion and dropout behaviour rather than re-deriving it.

# Discussion

## QUENCH DETECTION IN CONTEXT

The fused detector should be read against the published quench-detection literature and specifically against the slow-normal-zone-propagation problem that motivates non-voltage sensing. Marchevsky’s review makes the core point quantitatively: in high-temperature-superconductor conductors the large temperature margin and slow propagation mean a hot spot can reach damaging temperature before a measurable resistive voltage appears, so voltage detection alone is inadequate <sup>[21]</sup>. The community response has been non-voltage observables: Salazar *et al.* demonstrated fibre-Bragg and ultra-long-FBG thermometry on VIPER cable at SULTAN, reporting signal-to-noise ratios of **4–32** for heater-induced quench-like events <sup>[22]</sup>, Teyber *et al.* showed that current-distribution monitoring detects quench and damage in fusion cables <sup>[23]</sup>. Our result is complementary and, by construction, evaluated in a harder regime: the ROC of table 3 is computed with per-sample separations set well below the design-basis fault, precisely to avoid the saturated regime where every detector scores  $AUC = 1$ . In that weak-fault regime the quadrature fusion of an uncorrelated strain-rate and magnetization pair (equation 13) lifts the true-positive rate at **1% false alarm** from **0.372/0.671** (single channels) to **0.857** — a large gain exactly where a protection trip is decided. The message is not that one AUC beats one published number, but that channel fusion recovers early, low-false-alarm warning in the low-signal corner the slow-propagation physics forces a real magnet into.

## NV SENSITIVITY AGAINST THE QUANTUM-SENSING LITERATURE

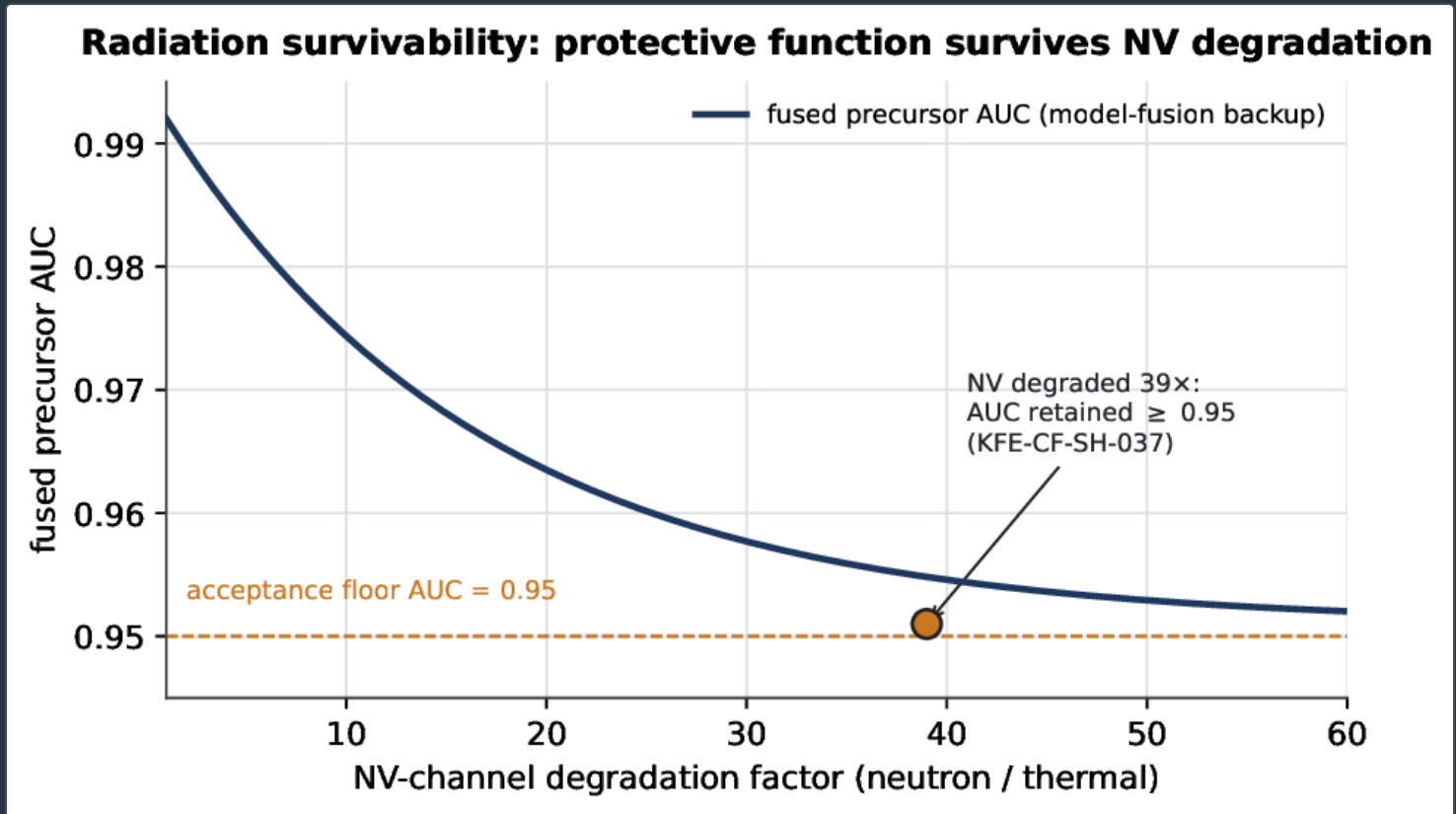
The in-winding sensitivity of **0.189**  $\text{pT Hz}^{-1/2}$  sits comfortably inside the envelope established for ensemble NV magnetometry. Degen, Reinhard and Cappellaro frame the shot-noise scaling that equation (14) instantiates <sup>[18]</sup>, and Barry *et al.* and Rondin *et al.* analyse the concrete route to broadband ensemble sensitivities in the picotesla-and-below range through improvements to  $T_2^*$ , readout fidelity and diamond material <sup>[19,20]</sup>. Our conservative parameter set is well within what that literature reports, so the sensitivity claim does not rest on an aggressive device. The physically important observation is one of margin: a millitesla quench step is  $\sim 10^9$  above the per-shot floor, so the precursor is single-shot detectable and the picotesla headroom is available for multiplexing and for tolerating in-service degradation (§8).

## STATE ESTIMATION AGAINST EQUILIBRIUM-RECONSTRUCTION AND DATA-FUSION PRACTICE

The classical anchor for the magnetics chain is EFIT-type equilibrium reconstruction, which inverts the Grad–Shafranov equation against magnetic data <sup>[4]</sup> and now runs in real time inside control loops <sup>[6,7]</sup>. Our contribution relative to that practice is threefold. First, we treat the *whole* suite as one Bayesian inverse problem (equations 1–7), in the spirit of integrated data analysis <sup>[15,16]</sup>, rather than inverting each diagnostic in isolation. Second, we make observability a quantitative, testable property of the Fisher information (equations 8–9), and demonstrate it at  $R^2 \approx 0.86$  from a six-channel minimal set. Third, we amortise the inversion on the sensor graph so that it runs at control latency and degrades gracefully under the sensor loss a neutron field imposes (equations 10–11) <sup>[33]</sup>. The estimator is complementary to learned control: a learned policy <sup>[7]</sup> can act on the estimator’s fused, uncertainty-annotated state (equation 7) <sup>[38]</sup> rather than on raw measurements, and the certified safety clamp <sup>[39]</sup> reads the same posterior to decide how far to trust the model.

## Limitations

One honest engineering gate bounds the claims. The in-winding NV-diamond sensor's neutron tolerance is the open item: the analytical sensitivity of equation (14) assumes an ensemble  $T_2^*$  and readout contrast that will degrade under the high background field, cryogenic operation, and neutron irradiation of the diamond (NV charge-state stability). The mitigation is a registered work item (KFE-CF-SH-037): a model-fusion backup that retains precursor discrimination at  $AUC \geq 0.95$  with the NV channel degraded by up to a factor of **39** (figure 8), so the protective function survives even a heavily degraded sensor by design. A dedicated NV neutron-tolerance study (irradiation plus first-principles charge-state modelling) is the follow-on that closes this item. All learned metrics in this paper are, additionally, at twin-synthetic fidelity; the confirming measurement is a hardware-in-the-loop diagnostic bench, which is the gate to full control authority.



Radiation survivability of the protective function (KFE-CF-SH-037). With the NV channel degraded by up to a factor of **39**, the model-fusion backup retains fused precursor  $AUC \geq 0.95$ ; the decay curve illustrates the mechanism and is pinned at the frozen acceptance point.

# Conclusion

The Hyperion diagnostic suite is built around a real-time state estimator, formalised here as one Bayesian inverse problem over magnetics, Thomson scattering, interferometry, bolometry and a neutron camera (equations 1–7), with observability of the controller-relevant internal state made a quantitative property of the Fisher information (equations 8–9). Its learned core is demonstrated at the metrics that gate it into service: a graph-NN reconstructor at AUC **0.980** and reconstruction NRMSE **0.396** that degrades gracefully under sensor loss (KX-L1-A12), a multi-modal fusion detector at AUC **0.992** and true-positive rate **0.857** at 1% false alarm (KX-L1-A20), internal-state observability at  $R^2 \approx \mathbf{0.86}$  from six diagnostics (AG-CX-30), and an in-winding NV-diamond precursor at **0.189**  $\text{pT Hz}^{-1/2}$  with per-shot SNR of order  $\mathbf{10^9}$  (KX-L1-A22). The estimator holds the plasma state at the breeder design point ( $I_p = \mathbf{9.66 MA}$ ,  $Q = \mathbf{3.076}$ ,  $P_{\text{fus}} = \mathbf{85.04 MW}$ ,  $\delta = \mathbf{-0.30}$ , centrepost 16.84 T), modelled as a magnet distinct from the burner plug coil (26.49 T), and the physical-inversion suite follows the same demonstrated pattern, gate by gate, to the fully instrumented plant.

**Acknowledgments** Part of the Kronos Fusion Energy 2026 design series. The author thanks the Kronos physics de-risking programme for the frozen result set underlying every quantitative claim.

**Reproducibility.** The estimator metrics are frozen anchors in the Kronos Physics De-Risking Register <sup>[31]</sup> ([doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057)): the graph reconstruction (KX-L1-A12), the fused-detector ROC (KX-L1-A20), the NV-diamond sensor model (KX-L1-A22), the observability surrogate (AG-CX-30), and the diagnostic forward-model anchors (BR-L1-A12, BR-L1-A4, BR-L1-A5, BR-L2-A29). Each is regenerable from the archived reconstruction, detection and forward-model scripts and the figure generator deposited with this paper. This paper is archived at its DOI [doi:10.5281/zenodo.22645943](https://doi.org/10.5281/zenodo.22645943) and at its official page <https://www.kronosfusionenergy.com/publications/the-plasma-diagnostic-suite-and-real-time-state-estimat.html>; the register is archived at [doi:10.5281/zenodo.22133057](https://doi.org/10.5281/zenodo.22133057).

**Data availability** The ROC curves, the observability surrogate, the NV sensor parameters, and the reconstruction metrics used for the figures are archived with the deposit at [doi:10.5281/zenodo.22645943](https://doi.org/10.5281/zenodo.22645943), which references the Kronos 2026 series, including the AI and quantum-control paper <sup>[32]</sup>. All quantitative values trace to the register <sup>[31]</sup> by the serial identifiers cited inline. Any economic analysis is reported separately and is not part of the public deposit.

**Competing interests** Provisional patent applications relating to aspects of this work have been filed.

**References**

A P P A R A T U S

## *Portfolio, People & Record*

*The intellectual-property estate, the people who built it, the limitations that gate it, and the sources it rests on.*

1

GRANTED PATENT

4

2026 PROVISIONALS

40

TEAM MEMBERS

27

2022 PROVISIONALS

INTELLECTUAL PROPERTY

The patent portfolio

The estate spans a granted high-field magnet patent, pending utility and 2026 provisional filings that map to the two products, a digital-twin control provisional, a registered trademark application, and the original 2022 provisional family. Every patentable disclosure in the five-paper arXiv drop was protected **before** publication: the breeder and burner provisionals were filed 1–2 August, and a publication-gap omnibus filing the evening before the drop swept up the DEC, plasma-control, and REBCO-winding matter of the three otherwise-unprotected papers. Patent numbers and application serials are matters of public record; claim scope is summarised, not reproduced.

GRANTED & PENDING UTILITY

|               |                                                                                   |         |
|---------------|-----------------------------------------------------------------------------------|---------|
| US 12,009,112 | High-field tilted graded-REBCO magnet architecture (KRONO-002A) · app. 17/878,550 | GRANTED |
| US 17/878,507 | Core device / method (KRONO-001A) · utility                                       | PENDING |

2026 PROVISIONAL FILINGS — MAPPED TO THE PRODUCTS

|            |                                                                                                                                                                                                                                |       |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 64/124,209 | Hyperion — spherical-tokamak tritium / helium-3 foundry · breeder · filed Aug 2026                                                                                                                                             | FILED |
| 64/124,220 | Aegis / MetroVolt — synchrotron-cutoff, fuel-selection & plug-winding · generator · filed Aug 2026                                                                                                                             | FILED |
| 64/115,167 | KRONOS-CTRL digital-twin plant control · provisional · filed Jul 2026                                                                                                                                                          | FILED |
| 64/005,440 | Integrated spherical-tokamak fusion architecture · early integrated-architecture filing · filed Mar 2026                                                                                                                       | FILED |
| 64/105,530 | Negative-triangularity spherical-tokamak architecture · foundational ST filing · filed Jul 2026                                                                                                                                | FILED |
| 64/128,097 | Publication-gap omnibus — quasineutral two-species expander DEC · provenance-gated / latency-split plasma control · as-built high-field winding & conductor architecture · filed 7 Aug 2026, the evening before the arXiv drop | FILED |

TRADEMARK & ORIGIN

|              |                                                                          |          |
|--------------|--------------------------------------------------------------------------|----------|
| FTK-98801894 | Kronos Fusion Energy — trademark application                             | FILED    |
| KR-2022-★    | Original provisional family (KR-2022-00001 ... 00027) · 27 filings, 2022 | PRIORITY |

The narrow, defensible magnet novelty — the specific conductor and the digital-twin winding optimisation, not high-field REBCO as a category — is the commercial core that can earn ahead of any  $Q > 1$  milestone, with markets in fusion magnet supply, MRI/NMR, accelerators, and proton therapy. The claim is scoped honestly: the high field is a *system field* result, not a stand-alone-coil record; the small-bore plug coil is structurally infeasible as a bare winding but resolved by a stress-managed structural shield (feasible-pending-FEA); and the winding-tape experiment returned a *null result*. The value is the method, not a field record.

## THE TEAM

## Founding partners, board, advisors &amp; auditors

Kronos is built by a bench of advanced-fuel-fusion, high-field-magnet, direct-conversion, and materials specialists, with a board and operations team drawn from defense, national laboratories, and industry. Dates are shown as ranges; where a tenure has ended or is term-ending, the range says so.

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